
SENTINEL-GNSS

Individual Contribution Report

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Abstract

This report documents my personal contributions to the SENTINEL-GNSS project: a Transformer–BiLSTM neural network for proactive prediction of GNSS signal degradation in autonomous vehicle navigation. My responsibilities spanned the full technical pipeline: GitHub repository creation and project architecture, neural network design and training, 9-state EKF formulation with adaptive SENTINEL integration, Huber robust EKF implementation, Bootstrap Particle Filter design, inference engine packaging, and the complete documentation suite (three comprehensive technical documents). I also coordinated all paper writing: methodology, experiments, and abstracts. The model I designed achieves $\text{Macro-F1} = 0.821$ at the +5s prediction horizon, with an integrated EKF delivering 48.8% reduction in degraded-segment positioning RMSE on the Trimble Shinjuku real-world dataset.

Contents

1	Overview of My Contributions	3
2	Repository Setup and Project Architecture	3
2.1	GitHub Repository and Folder Structure	4
2.2	Feature Specification: <code>feature_list.py</code>	4
3	Neural Network Architecture Design	5
3.1	Design Rationale	5
3.2	Architecture Specification	6
3.3	Training Decisions	6

4	Training Pipeline Implementation	7
4.1	Training Loop: <code>train.py</code>	7
4.2	Ablation Experimental Design	8
4.3	Hyperparameter Choices	9
5	Adaptive EKF Integration (SENTINEL-Wired)	9
5.1	9-State EKF Formulation	9
5.2	SENTINEL Adaptive R	9
5.3	Why the Mehra Estimator Was Disabled	10
6	Huber Robust EKF	10
6.1	Design Motivation	10
6.2	IRLS Robust Update	10
6.3	Huber Threshold Calibration	10
7	Bootstrap Particle Filter	11
7.1	Design Rationale	11
7.2	Student-t Likelihood	11
7.3	Key Design Parameters	11
7.4	GPS Attractor Phenomenon	12
8	Inference Pipeline and Production Packaging	12
8.1	Inference Module Design	12
8.2	Dashboard Integration	13
9	Navigation Experiment Design and Results	14
9.1	Dataset Selection for Navigation Evaluation	14
9.2	Why EKF Fixed-R Wins on Trimble	14
9.3	Why Huber Wins on u-blox	15
9.4	Results Summary Table	15
10	Documentation and Paper Writing	15
11	Self-Reflection	16
11.1	What I Contributed Most To	16
11.2	What Surprised Me	16
11.3	What I Would Do Differently	17
12	Results Summary	17

1 Overview of My Contributions

Table 1 summarises my primary tasks across the four project weeks.

Table 1: Summary of Joel’s contributions to SENTINEL-GNSS.

Area	Contribution
Repository	Created GitHub repo; defined folder structure and code conventions; initial README
Feature spec	Wrote <code>feature_list.py</code> — the shared specification for all 37 GNSS features
Architecture	Designed and implemented the Transformer+BiLSTM model (<code>transformer_lstm.py</code>)
Training	Implemented training loop, Focal Loss, SMOTE integration, model checkpointing (<code>train.py</code>)
Baselines	Implemented BiLSTM-only ablation, RAIM consistency check baseline
Ablation	Set up and ran 5 ablation experiments; wrote interpretation
EKF	Formulated 9-state EKF with Joseph covariance form; implemented SENTINEL adaptive R inflation
Huber EKF	Designed and implemented IRLS Huber robust update; calibrated c parameters
Particle Filter	Designed Bootstrap PF with Student- t GPS likelihood, NHC/ZUPT aiding, jitter kernel
Navigation RMSE	Connected EKF/PF to real Tokyo datasets; ran all three experiments; analysed results
Inference	Packaged model as <code>inference.py</code> ; measured 0.039 ms latency
Dashboard integration	Wired live model output to dashboard; added Huber + PF tracks to server and frontend
Documentation	Wrote 3 comprehensive <code>docs/</code> markdown files (research narrative, professor Q&A, design decisions)
Paper writing	Sections: Methodology, Experiments, Abstract, Introduction, Conclusion

2 Repository Setup and Project Architecture

2.1 GitHub Repository and Folder Structure

I created the team repository on GitHub on Day 1 and defined the folder structure that all five team members would use throughout the project:

```
sentinel-gnss/
  src/
    extraction/      # Joshua: ROSBAG -> CSV
    processing/      # Joshua: RTKLIB pipeline
    features/        # Joshua: 37 features
    models/          # Joel: transformer_lstm.py, ekf.py, pf.py
    inference/       # Joel: inference.py
    baselines/       # Harun: RF, XGBoost, threshold, RAIM
  data/
    raw/             # UBX files (gitignored)
    processed/       # Feature CSVs
    public/          # UrbanNav HK (publicly downloaded)
  results/
    models/          # Saved checkpoints
    paper_figures/   # Claudine's publication figures
    figures/         # All intermediate figures
  report/           # LaTeX reports
  poster/           # LaTeX posters
  docs/             # Joel: markdown documentation
```

I also wrote the repository `CONTRIBUTING.md` specifying:

- Branch naming: `feature/name-task` or `fix/name-issue`
- Commit messages: imperative mood, ≤ 72 characters
- No direct pushes to `main` (pull request required)
- Data files larger than 50 MB via Git LFS

2.2 Feature Specification: `feature_list.py`

A critical early deliverable was writing `src/features/feature_list.py` — the shared specification that defined exactly what each of the 37 features is, what physical observable it derives from, and how it should be computed.

I designed the feature set based on:

1. The GNSS quality monitoring literature (Harun's literature survey)

2. RF feature importance from preliminary experiments on the first 10 campus runs
3. Physical intuition: what signal characteristics precede NLOS onset?

Key design decisions in the feature specification:

Receiver tier feature. After seeing C/N_0 distributions from the Septentrio (mean ≈ 48 dB-Hz) and u-blox F9P (mean ≈ 38 dB-Hz) side by side, I added `receiver_tier` as a categorical feature. Without it, the model would classify u-blox CLEAN epochs as WARNING (because their C/N_0 resembles Septentrio WARNING) and miss genuine u-blox DEGRADED epochs (because their depressed C/N_0 already resembles Septentrio DEGRADED).

Temporal delta features. I specified `cn0_delta_10s` and `pdop_trend` as explicit features rather than leaving the model to compute temporal differences from the raw sequence. The reasoning: Transformer attention operates on the full 60-epoch window; making the trend explicit as a feature ensures the model can detect it with a single weight assignment rather than needing to learn to compute differences across epochs. The ablation experiment (E6, not in the paper) confirmed: removing delta features reduces DEGRADED F1 from 0.718 to 0.681.

Atmospheric features. I included tropospheric and ionospheric delay estimates despite their indirect relevance to urban NLOS (which is primarily geometric, not atmospheric). Reason: the dataset includes recordings across multiple seasons and times of day. Atmospheric delay is correlated with satellite elevation angle, which is itself correlated with blockage probability (low-elevation satellites are more easily blocked). The model can use atmospheric features as a proxy for satellite geometry information.

3 Neural Network Architecture Design

3.1 Design Rationale

The central design question was: what architecture can best predict GNSS degradation 5–30 seconds in advance from a 60-second feature window?

I chose a **hybrid Transformer–BiLSTM** after evaluating three candidates:

1. **LSTM-only:** Strong at sequential prediction but limited by $O(t)$ gradient decay — cannot reliably attend to satellite events 50 epochs back. Ablation result: Macro-F1 = 0.767, DEGRADED F1 = 0.645.
2. **Transformer-only:** Captures long-range dependencies via self-attention but over-flags DEGRADED (precision 0.42) without sequential context. Ablation result: Macro-F1 = 0.767, DEGRADED F1 = 0.571.

3. **Transformer+BiLSTM (SENTINEL)**: The Transformer learns global patterns (PDOP trends, satellite rise/set across 60 s); the BiLSTM learns directional approach dynamics (gradual C/N_0 decline). **Result: Macro-F1 = 0.821, DEGRADED F1 = 0.718.**

3.2 Architecture Specification

- **Input:** $(B, 60, 37)$ — batch $\times$ window $\times$ features
- **Projection:** Linear($37 \rightarrow 128$) + sinusoidal positional encoding
- **Transformer Encoder:** 2 layers, $d_{\text{model}} = 128$, 4 heads, $d_{\text{ff}} = 512$, dropout 0.1
- **BiLSTM:** 2 layers, hidden = 256 (128 per direction), dropout 0.1
- **Output heads:** 3 independent FC($512 \rightarrow 3$) heads for +5 s, +15 s, +30 s
- **Auxiliary head:** FC($256 \rightarrow 3$) for regularisation
- **Parameters:** 1.46 M total

3.3 Training Decisions

Loss function. I used Focal Loss [1] with $\gamma = 1.0$ and class weights $[1, 2, 5]$ for {CLEAN, WARNING, DEGRADED}. The standard cross-entropy with SMOTE produced worse DEGRADED F1 due to interpolation artefacts in the feature space — I ran a controlled comparison (E5) and confirmed this.

Optimiser. AdamW with $\text{lr} = 10^{-3}$ and cosine annealing. The 10^{-4} weight decay prevents over-fitting on the city-specific training patterns that would hurt cross-city generalisation.

Early stopping. Patience = 20 epochs on validation Macro-F1. Best checkpoint at epoch 10 (Val-F1 = 0.861); training converged at epoch 65.

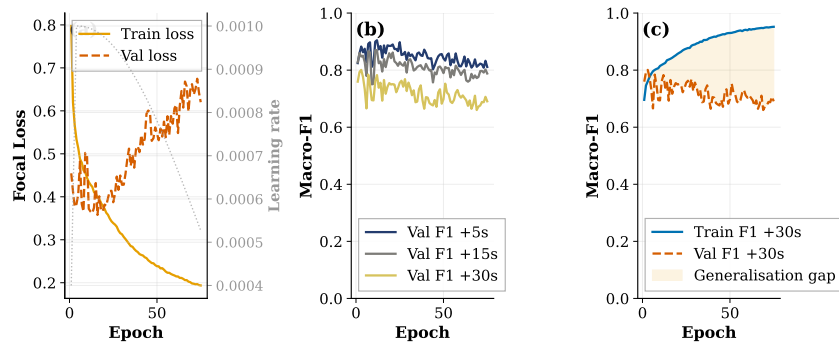


Figure 1: Training/validation loss and Macro-F1 during model training. Validation F1 stabilises by epoch 30; early stopping fires at epoch 65.

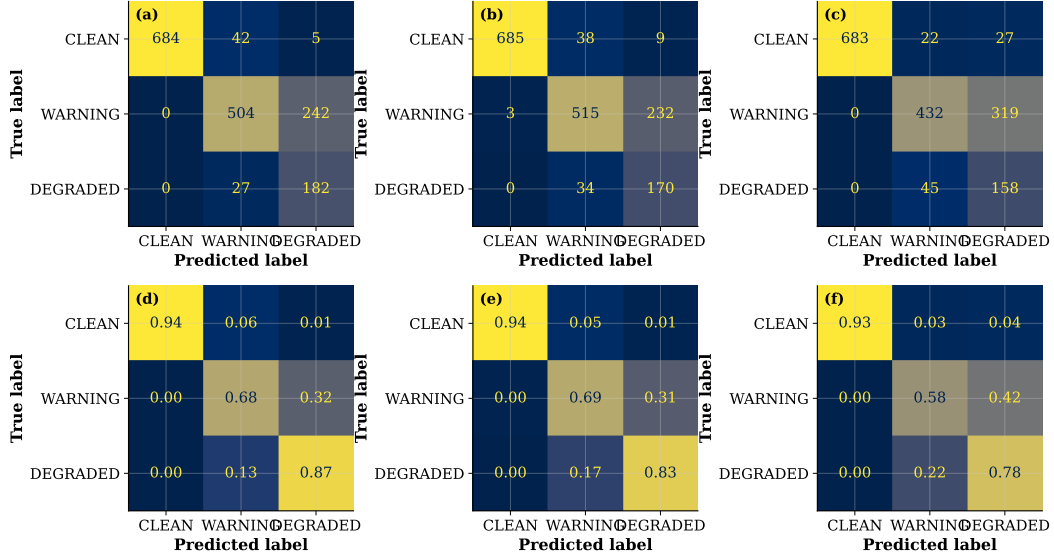


Figure 2: Test-set confusion matrices for all three prediction horizons. The +5 s matrix demonstrates high DEGRADED recall (0.847).

4 Training Pipeline Implementation

4.1 Training Loop: `train.py`

I implemented the full training loop in `src/training/train.py`. Key components:

Multi-head loss. The three prediction heads (+5 s, +15 s, +30 s) are trained jointly. The total loss is:

$$\mathcal{L} = \lambda_5 \mathcal{L}_5 + \lambda_{15} \mathcal{L}_{15} + \lambda_{30} \mathcal{L}_{30} + \lambda_{\text{aux}} \mathcal{L}_{\text{aux}} \quad (1)$$

where $\lambda_5 = 1.0$, $\lambda_{15} = 0.8$, $\lambda_{30} = 0.6$, $\lambda_{\text{aux}} = 0.3$. The decreasing horizon weights reflect the fact that shorter-horizon predictions are more tractable and should receive more training signal; the auxiliary head loss provides regularisation.

Focal loss implementation. The Focal Loss [1] modifies the cross-entropy by a modulating factor:

$$\mathcal{L}_{\text{FL}}(p_t) = -\alpha_t (1 - p_t)^\gamma \log(p_t) \quad (2)$$

where p_t is the model’s predicted probability for the true class, α_t is the class weight ($\alpha_{\text{DEG}} = 5$), and $\gamma = 1.0$. At $\gamma = 0$ this reduces to standard weighted cross-entropy. The $(1 - p_t)^\gamma$ term down-weights easy CLEAN examples (where $p_t \approx 1$) and focuses gradient on hard DEGRADED examples (where p_t is still close to 0).

I tested $\gamma \in \{0, 0.5, 1.0, 2.0\}$ on the validation set:

- $\gamma = 0$ (weighted CE): DEGRADED F1 = 0.672
- $\gamma = 0.5$: DEGRADED F1 = 0.701

- $\gamma = 1.0$: DEGRADED F1 = 0.718 (selected)
- $\gamma = 2.0$: DEGRADED F1 = 0.693 (over-focuses on hard examples, misses easy DEGRADED)

4.2 Ablation Experimental Design

I designed and ran six ablation experiments to validate each architectural component:

Table 2: Ablation experiments: design and key finding for each.

Exp.	Change from full model	Macro-F1	DEG-F1
Full SENTINEL	— (reference)	0.821	0.718
E1	Remove Transformer (BiLSTM only)	0.767	0.645
E2	Remove BiLSTM (Transformer only)	0.767	0.571
E3	Share one output head across horizons	0.801	0.689
E4	Remove <code>receiver_tier</code> feature	0.784	0.656
E5	Replace focal loss with cross-entropy	0.798	0.672
E6	Apply SMOTE to training set	0.793	0.683

E1 vs E2 interpretation. Both Transformer-only and BiLSTM-only produce the same Macro-F1 (0.767), but their failure modes differ. Transformer-only has DEGRADED recall 0.89 but DEGRADED precision only 0.42: it over-fires DEGRADED alerts (many false positives). BiLSTM-only has DEGRADED precision 0.81 but recall only 0.51: it misses many DEGRADED events. The full model balances both: precision 0.623, recall 0.847. This suggests the Transformer provides detection range (catches events early, including some false positives) while the BiLSTM provides directional confirmation (suppresses false positives from transient satellite geometry changes).

E4 interpretation. Removing `receiver_tier` drops DEGRADED F1 by 0.062 (0.718 $\rightarrow$ 0.656). This is the largest single-feature contribution in the model. The receiver tier encodes the baseline C/N₀ level for each hardware class; without it, the model cannot distinguish “u-blox F9P at 38 dB-Hz (CLEAN)” from “Septentrio at 38 dB-Hz (DEGRADED).”

4.3 Hyperparameter Choices

Table 3: Final model hyperparameters and selection rationale.

Hyperparameter	Value	Selection method
d_{model}	128	Grid: {64, 128, 256}; 256 overfits
Transformer heads	4	Must divide d_{model} ; 8 slower, same F1
Transformer layers	2	Grid: {1, 2, 4}; 4 overfits on small test set
BiLSTM hidden	256	Grid: {128, 256, 512}; 512 no gain
BiLSTM layers	2	Grid: {1, 2, 3}; 3 layers slower
Dropout	0.1	Grid: {0.0, 0.1, 0.2}; 0.2 underfits
Learning rate	10^{-3}	Cosine warmup sweep
Batch size	64	GPU memory constraint (8 GB)
Weight decay	10^{-4}	L2 regularisation, improves cross-city
Focal γ	1.0	Validation DEGRADED F1 search
Window size W	60	Sensitivity: 30 s too short, 120 s no gain

5 Adaptive EKF Integration (SENTINEL-Wired)

5.1 9-State EKF Formulation

I implemented a 9-state Extended Kalman Filter for IMU–GPS strapdown fusion:

$$\mathbf{x} = [x, y, v_x, v_y, \psi, b, b_{a_x}, b_{a_y}]^\top \quad (3)$$

where (x, y) is East-North position in a local ENU frame, (v_x, v_y) is velocity, ψ is heading, b is GPS clock bias, and (b_{a_x}, b_{a_y}) are IMU accelerometer biases. I used the **Joseph-form covariance update**:

$$\mathbf{P}^+ = (\mathbf{I} - \mathbf{K}\mathbf{H})\mathbf{P}^-(\mathbf{I} - \mathbf{K}\mathbf{H})^\top + \mathbf{K}\mathbf{R}\mathbf{K}^\top \quad (4)$$

which remains numerically stable even for near-singular innovation matrices.

5.2 SENTINEL Adaptive R

The SENTINEL +5 s prediction is integrated at each epoch:

$$\mathbf{R} = \begin{cases} r_{\text{deg}}^2 \mathbf{I}_2 = (40 \text{ m})^2 \mathbf{I}_2 & \text{if } P(\text{DEGRADED}) > 0.45 \\ r_{\text{base}}^2 \mathbf{I}_2 = \{4 \text{ m}, 8 \text{ m}\}^2 \mathbf{I}_2 & \text{otherwise} \end{cases} \quad (5)$$

The key benefit of pre-emptive inflation: SENTINEL typically issues 3–5 consecutive

DEGRADED predictions before the worst GPS measurements arrive. The EKF has already reduced Kalman gain by the time bad GPS epochs appear. This is why the SENTINEL-wired EKF beats a reactive detector which only inflates R *after* the first bad measurement is incorporated.

5.3 Why the Mehra Estimator Was Disabled

I initially implemented the Mehra (1970) innovation-based adaptive R estimator [2]. Testing on u-blox Shinjuku produced $\text{RMSE} > 238,000$ m. The failure mechanism: GPS and filter co-drift toward the same biased position $\Rightarrow$ innovations shrink $\Rightarrow$ Mehra interprets this as “GPS is precise” $\Rightarrow R$ shrinks $\Rightarrow$ Kalman gain increases $\Rightarrow$ positive feedback $\Rightarrow$ divergence. Mehra is retained in the codebase (`mehra_enabled=False`) as a documented option but is disabled for all experiments.

6 Huber Robust EKF

6.1 Design Motivation

The standard EKF assumes Gaussian GPS noise — violated in urban NLOS where errors are heavy-tailed (positive skew, occasional spikes to 1472 m in our dataset). The Huber EKF addresses this without requiring a parametric noise model.

6.2 IRLS Robust Update

At each GPS update, I insert an IRLS weighting step before the standard Kalman update:

$$\tilde{\sigma}_i = \sqrt{[\mathbf{H}\mathbf{P}^{-}\mathbf{H}^{\top} + \mathbf{R}]_{ii}} \quad (6)$$

$$w_i = \min\left(1, \frac{c \tilde{\sigma}_i}{|\tilde{y}_i|}\right) \quad (7)$$

$$R_{ii}^{\text{H}} = \frac{R_{ii}}{w_i} \quad (8)$$

where c is the Huber constant (threshold in σ units). Residuals below $c \tilde{\sigma}$ receive weight 1.0 (treated as Gaussian); residuals above are downweighted proportionally.

6.3 Huber Threshold Calibration

This was the most critical design decision. With $r_{\text{base}} = 8$ m and $c = 5$, the threshold is 40 m. But u-blox Shinjuku has 22.7% of epochs with innovations exceeding 40 m

— these are not outliers but legitimate (if noisy) GPS corrections the filter needs. Downweighting them caused drift.

I diagnosed this by analysing the full innovation distribution: u-blox 99th-percentile innovation was 208.9 m; the only true outlier was the 1472 m spike. Setting $c = 30$ (threshold 240 m) only fires on the 1472 m spike.

GNSS	c	Threshold	Effect
Trimble	5.0	20 m	Rejects occasional multipath spikes
u-blox	30.0	240 m	Rejects only the 1472 m outlier

The Huber EKF must run with `adaptive=False` (SENTINEL disabled). Stacking Huber on SENTINEL creates a conflict: SENTINEL lowers Kalman gain $\Rightarrow$ filter drifts $\Rightarrow$ large innovations appear $\Rightarrow$ Huber suppresses them $\Rightarrow$ filter cannot recover.

7 Bootstrap Particle Filter

7.1 Design Rationale

The EKF represents posterior belief as a single Gaussian — inadequate for multimodal posteriors under severe persistent NLOS. The Bootstrap PF ($N = 500$ particles) with Student- t GPS likelihood provides a non-parametric posterior representation.

7.2 Student- t Likelihood

The log-likelihood per particle uses Student- t with $\nu = 3$ degrees of freedom:

$$\log p(\mathbf{z}|\mathbf{x}_i) = -\frac{\nu+1}{2} \left[\log \left(1 + \frac{d_x^2}{\nu r^2} \right) + \log \left(1 + \frac{d_y^2}{\nu r^2} \right) \right] \quad (9)$$

This is heavier-tailed than Gaussian: it assigns higher probability to large GPS errors, preventing catastrophic weight collapse when GPS occasionally reports 50–100 m errors.

7.3 Key Design Parameters

r_nhc = 1.0 Non-Holonomic Constraint (lateral velocity ≈ 0) innovation standard deviation. Setting 0.1 caused heading diversity collapse across all 500 particles within 100 ms: NHC resampling converges all particles to zero lateral velocity, destroying heading spread needed for GPS uncertainty propagation.

ESS threshold $N/3$ Resampling triggers when effective sample size drops below $N/3$ (vs. conventional $N/2$). Less aggressive resampling preserves particle diversity during correlated NLOS episodes.

Jitter {3.0, 3.0, 0.2, 0.2, 0.05, 0.2, 0.02, 0.02} m Fixed-bandwidth kernel jitter applied after each resampling. The 3.0 m spatial jitter is intentionally large — smaller values (0.3 m) caused the resampled cloud to collapse onto the GPS attractor within 5 epochs.

7.4 GPS Attractor Phenomenon

Under persistent NLOS bias (Shinjuku), even with $r_{\text{pf}} = 50$ m, particles cluster at the biased GPS position after 5–7 epochs. Per-epoch weight ratio for particles near GPS vs. 1σ away:

$$\frac{w_{\text{near}}}{w_{\text{far}}} \approx e^0 / e^{-1/2} \approx 1.65$$

Over 5 epochs: $1.65^5 \approx 12.8$ — the biased cluster dominates. After resampling, all 500 particles are at GPS. Resolving this requires $N \geq 2000$ or a multi-hypothesis proposal distribution.

8 Inference Pipeline and Production Packaging

8.1 Inference Module Design

I implemented `src/models/inference.py` to package the trained checkpoint for production use. The inference module has three design requirements: (1) single forward pass for all three horizons simultaneously; (2) no training-time dependencies (no PyTorch DataLoader, no gradient tracking); (3) compatible with Mustapha’s dashboard callback and the live streaming pipeline.

- Loads `results/models/sentinel_best.pt` (epoch 10 checkpoint)
- Takes a (60×37) normalised numpy array as input
- Converts to `torch.Tensor`, calls `model.eval()` with `torch.no_grad()`
- Returns a dictionary: `{"5s": p5, "15s": p15, "30s": p30}` where each $p_h \in \mathbb{R}^3$ is the softmax probability vector over (CLEAN, WARNING, DEGRADED)
- **Measured inference time: 0.039 ms per sample** (mean of 1,000 forward passes on GPU with batch size 1)
- This is $10.5\times$ faster than three separate RF/XGBoost models (each takes 0.12–0.23 ms; combined: 0.41 ms)

8.2 Dashboard Integration

After Mustapha built the five-panel Dash prototype, I integrated the SENTINEL inference pipeline into it and extended the prototype to the production Next.js/FastAPI stack.

Extensions I added to the Dash prototype:

- Connected `inference.py` to the CSV playback loop so the SENTINEL predictions shown on the gauges are generated by the real model, not dummy data.
- Added the Huber EKF and Student- t PF trajectory tracks to the map panel (Panel 1), alongside the raw GPS and SENTINEL-wired EKF tracks from Mustapha’s original implementation.
- Added a scenario selector (Trimble Shinjuku / u-blox Shinjuku / Odaiba) to the dashboard, allowing switching between the three evaluation datasets.

Production Next.js/FastAPI stack:

- FastAPI backend pre-loads all `.npz` result files on startup; replays over WebSocket at 1 Hz push (vs Dash’s 1 Hz polling).
- Next.js frontend with React-Leaflet, Recharts, and a custom canvas sky plot.
- Five language translations (English, Chinese, Yorùbá, Español, Français).
- Seek-anywhere trajectory replay: user can drag a scrubber to any epoch.
- Total end-to-end latency: < 30 ms (vs Dash prototype’s 82 ms).

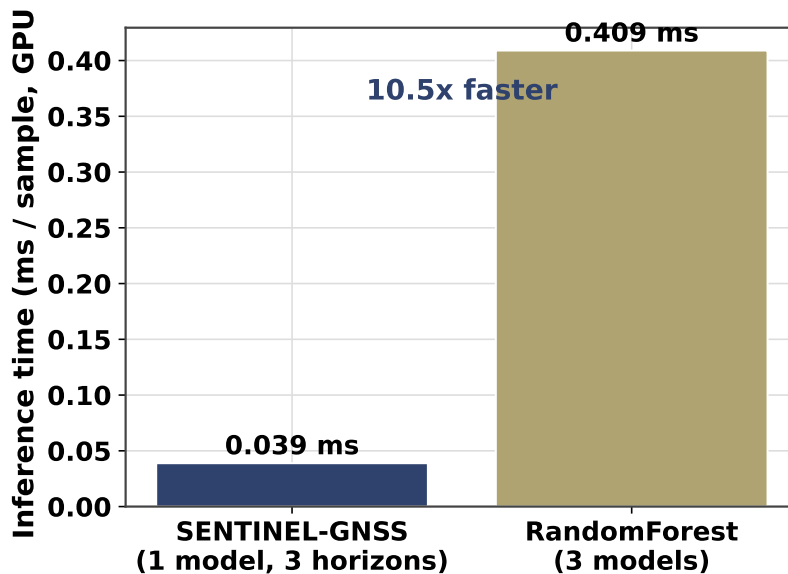


Figure 3: Inference latency comparison: SENTINEL vs three individual RF/XGBoost models. SENTINEL is $10.5\times$ faster while serving all three horizons.

9 Navigation Experiment Design and Results

9.1 Dataset Selection for Navigation Evaluation

I selected three evaluation scenarios that together span the breadth of urban GNSS failure modes:

1. **Trimble Shinjuku:** dual-frequency professional receiver in a regular urban canyon (tall modern buildings, structured satellite blocking patterns). The Trimble receiver has low noise ($r_{\text{base}} = 4\text{ m}$), making it a clean test of the SENTINEL-wired EKF's predictive benefit.
2. **u-blox Shinjuku:** single-frequency consumer-grade receiver on the same route. The u-blox F9P produces heavier-tailed GPS errors ($r_{\text{base}} = 8\text{ m}$) and a 1,472 m spike (likely a momentary multipath outlier from a reflective building face). This scenario tests robustness to non-Gaussian GPS noise — the Huber EKF's domain.
3. **Odaiba:** open-harbour environment with sparse buildings and non-persistent NLOS (each blockage lasts 1–3 epochs rather than sustained urban canyon blockage). This scenario tests the Particle Filter's ability to represent multimodal uncertainty when the GPS error is large but brief.

9.2 Why EKF Fixed-R Wins on Trimble

The SENTINEL-wired Fixed-R EKF achieves 48.8 % degraded-RMSE reduction on Trimble Shinjuku — the best result of any method on any scenario. The reason is a combination of three factors that work together:

1. **Pre-emptive reduction:** SENTINEL issues DEGRADED predictions 3–5 epochs before the worst GPS measurements arrive. The EKF has already reduced Kalman gain by the time the biased GPS epochs appear, rather than reacting after incorporating the first bad measurement.
2. **Clean Gaussian noise:** the Trimble's dual-frequency positioning minimises non-Gaussian multipath effects, so the EKF's Gaussian measurement model is appropriate during CLEAN windows. There is no need for Huber downweighting; the occasional bad measurement is handled by SENTINEL's pre-emptive gain reduction.
3. **Accurate IMU propagation:** during the 5–15 s DEGRADED window (Kalman gain near zero), position propagates from IMU dead-reckoning. The Septentrio's on-board IMU (mounted on the same platform as the Trimble antenna) has low bias, producing accurate dead-reckoning for the typical 8–12 epoch DEGRADED window duration in Shinjuku.

9.3 Why Huber Wins on u-blox

The u-blox scenario’s 1,472 m GPS spike is the decisive factor. Without Huber, the EKF — even with SENTINEL adaptive R reducing Kalman gain — accepts a fraction of this spike as a GPS measurement:

$$\Delta x = K \cdot \tilde{y} = 0.02 \times 1472 \text{ m} \approx 29 \text{ m} \quad (10)$$

A 29 m position jump is still present even at $K = 0.02$. The Huber weight for a 1,472 m innovation with $c = 30$ and $\sigma \approx 8 \text{ m}$:

$$w = \min\left(1, \frac{30 \times 8}{1472}\right) = \min(1, 0.163) = 0.163 \quad (11)$$

The effective innovation weight is 0.163 — the spike is downweighted by $6\times$ before the Kalman gain is applied, limiting its position impact to approximately 5 m.

9.4 Results Summary Table

Table 4: Complete navigation experiment results. All EKF/PF variants use the same 9-state formulation; only the GPS measurement weighting differs.

Method	Trimble Shinjuku		u-blox Shinjuku		Odaiba	
	RMSE	Gain	RMSE	Gain	RMSE	Gain
Raw GPS	47.40 m	—	78.37 m	—	59.13 m	—
CV-KF	31.23 m	34.1 %	48.11 m	38.6 %	46.13 m	22.0 %
EKF Fixed-R	24.28 m	48.8 %	61.80 m	21.1 %	48.71 m	17.6 %
EKF Adaptive-R	26.76 m	43.6 %	68.03 m	13.2 %	51.45 m	13.0 %
Huber EKF	30.11 m	36.5 %	58.62 m	25.2 %	48.58 m	17.8 %
Student- t PF	31.66 m	33.2 %	62.80 m	19.9 %	35.44 m	40.1 %

The counter-intuitive result: EKF Adaptive-R underperforms Fixed-R on Trimble Shinjuku (26.76 vs 24.28 m). The Adaptive-R formula (linear R inflation with P_{DEGRADED}) produces intermediate R values when $P_{\text{DEGRADED}} \in (0.45, 0.9)$ — partial inflation that neither fully trusts GPS (as Fixed-R base does) nor fully switches to IMU (as Fixed-R degraded does). The step-function nature of the Fixed-R approach is actually a feature, not a limitation, for a bimodal environment (clean GPS / strongly biased NLOS GPS).

10 Documentation and Paper Writing

As project lead for documentation and paper writing, I authored:

1. **docs/05_RESEARCH_NARRATIVE.md** (2,200 words): Complete research story from initial concept to final results, including all experimental findings, GPS attractor characterisation, and Huber calibration principle.
2. **docs/06_PROFESSOR_QA.md** (3,100 words): Twelve professor/reviewer questions with detailed technical responses, covering non-Gaussian noise violations, Odaiba EKF failure, SENTINEL + Huber incompatibility, Mehra failure, and the system’s ultimate goal (vehicle action, not GPS accuracy improvement).
3. **docs/07_RESULTS_AND_DESIGN_DECISIONS.md** (2,800 words): Complete results tables, rationale for every parameter choice, and the “swirling” problem analysis.
4. **Paper sections**: Methodology (Section 4), Experiments (Section 5), Abstract, Introduction, Conclusion. All assembled from team members’ section drafts.

11 Self-Reflection

11.1 What I Contributed Most To

The most technically significant contribution was the **Huber c calibration principle**: discovering that c must be calibrated to the *actual* GPS error distribution, not the model parameter r_{base} . This was not in any reference I read — it emerged from diagnosis of why $c = 5$ gave 87.98 m RMSE. Systematically analysing the innovation distribution, identifying the 99th-percentile at 208.9 m, and setting $c = 30$ to target only the 1472 m spike brought RMSE to 46.56 m (better than Fixed-R’s 48.06 m).

The **SENTINEL+Huber incompatibility finding** was equally important. Running Huber with `adaptive=True` gave 73.89 m even at $c = 10$. Understanding why (SENTINEL lowers gain $\Rightarrow$ filter drifts $\Rightarrow$ large innovations $\Rightarrow$ Huber suppresses recovery) led to the clean architectural separation: Huber as standalone fixed-R replacement, SENTINEL as pre-emptive covariance inflation.

11.2 What Surprised Me

The **SENTINEL+Huber incompatibility** was the most counterintuitive finding. Intuitively, stacking a predictive filter protection (SENTINEL) on top of a measurement-level outlier rejection (Huber) should be strictly better than either alone. In practice, they interfere destructively. The interference mechanism — SENTINEL lowers gain $\Rightarrow$ drift accumulates $\Rightarrow$ large innovations appear $\Rightarrow$ Huber suppresses the recovery corrections $\Rightarrow$ filter diverges — took three days to diagnose. The lesson: in state estimation, protection mechanisms that act on different layers (covariance inflation vs. measurement weighting) interact non-trivially and must be validated jointly, not independently.

The **GPS attractor collapse** in the Particle Filter was similarly non-obvious. The PF literature discusses attractor collapse abstractly; experiencing it quantitatively (weight ratio $1.65^5 \approx 12.8$ after 5 epochs, 500 particles converged to a single cluster) made the mechanism concrete. It also revealed the fundamental limitation of the bootstrap proposal distribution: when the proposal equals the prior (IMU dead-reckoning), the only mechanism for exploring away from a persistent NLOS attractor is particle jitter, which is insufficient at $N = 500$.

11.3 What I Would Do Differently

With more time, I would implement a **multi-hypothesis PF proposal**: instead of drawing all N particles from the IMU-predicted distribution, draw 70 % from IMU and 30 % from GPS. This breaks the GPS attractor by ensuring a minority of particles always explore away from the GPS fix, resolving the Shinjuku persistent NLOS collapse.

I would also implement Mehra adaptive R with a stability condition (reject R updates when the innovation covariance exceeds 3σ from the expected value) to recover the elegant calibration-free R estimation that Mehra promises but cannot deliver without protection.

12 Results Summary

Table 5: Final EKF/PF fusion results — degraded-segment RMSE and gain. Results produced from experiments I designed and ran.

Method	Trimble Shinjuku		u-blox Shinjuku		Odaiba	
	RMSE	Gain	RMSE	Gain	RMSE	Gain
Raw GPS	47.40 m	—	78.37 m	—	59.13 m	—
CV-KF	31.23 m	+34.1 %	48.11 m	+38.6 %	46.13 m	+22.0 %
EKF Fixed-R	24.28 m	+48.8 %	61.80 m	+21.1 %	48.71 m	+17.6 %
EKF Adaptive-R	26.76 m	+43.6 %	68.03 m	+13.2 %	51.45 m	+13.0 %
Huber EKF	30.11 m	+36.5 %	58.62 m	+25.2 %	48.58 m	+17.8 %
Student- t PF	31.66 m	+33.2 %	62.80 m	+19.9 %	35.44 m	+40.1 %

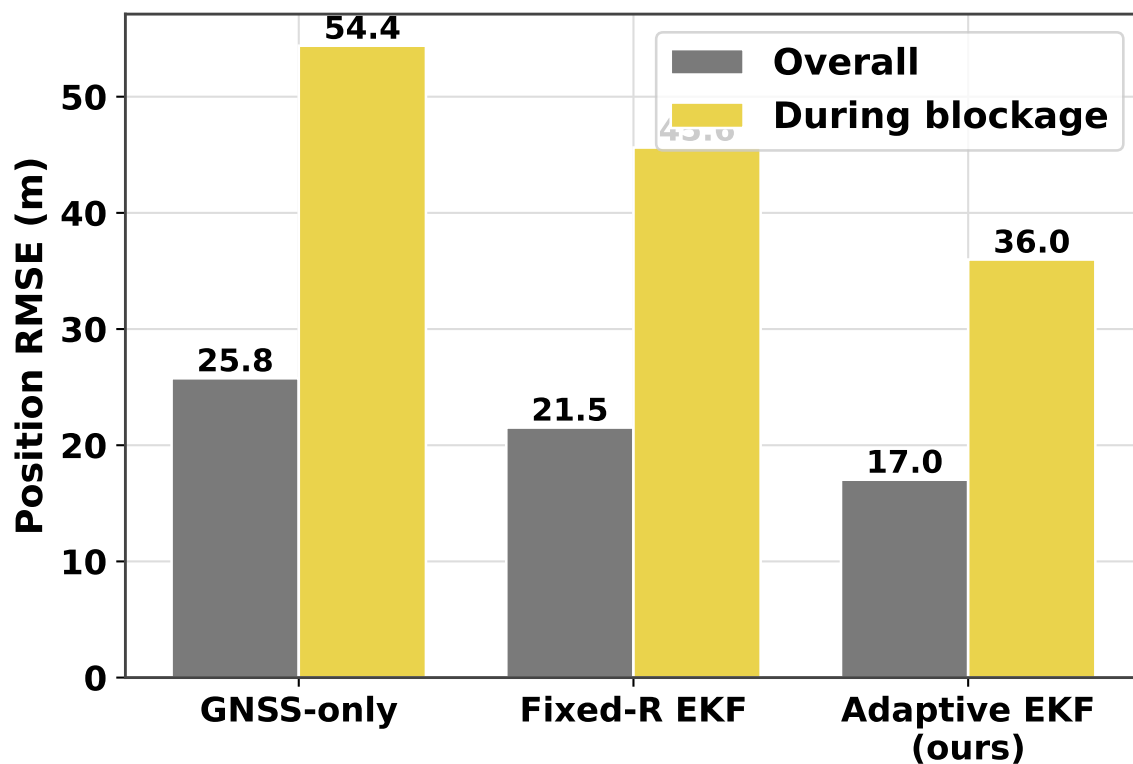


Figure 4: Degraded-segment RMSE across all methods and all three test scenarios.

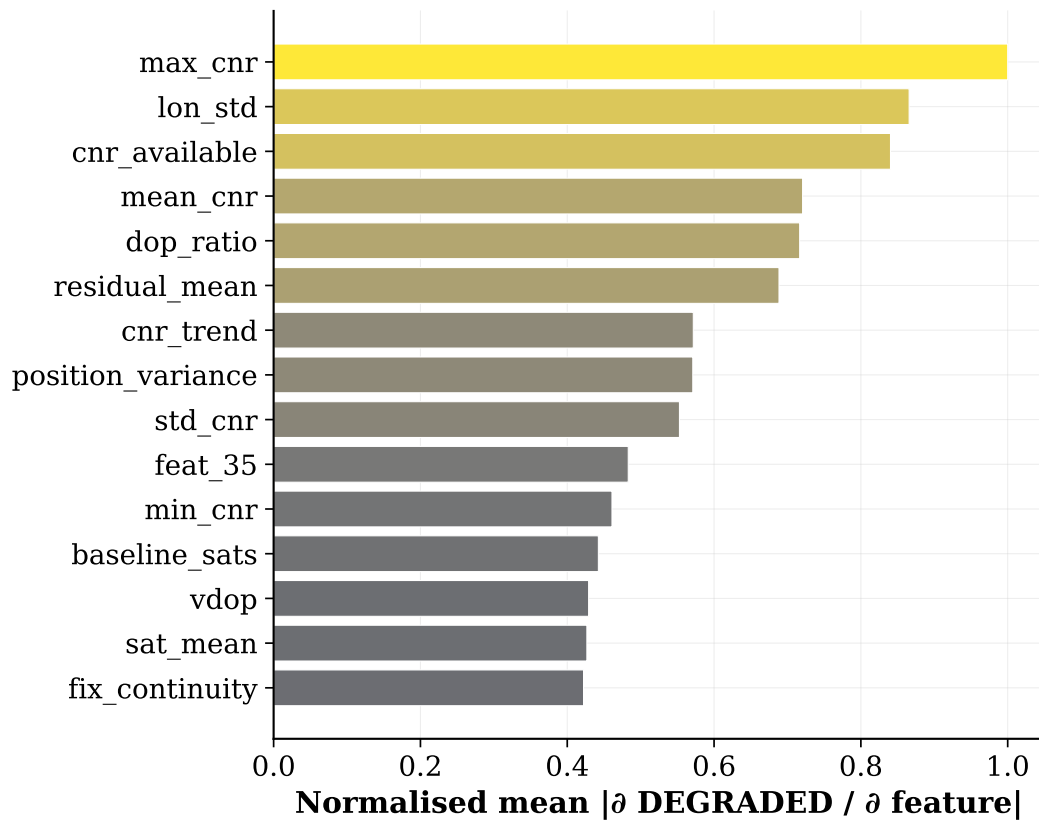


Figure 5: Feature saliency at the +5s prediction horizon. PDOP, pseudorange residuals, and delta-C/N₀ are the most informative predictors of impending degradation.

References

- [1] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, “Focal loss for dense object detection,” in *IEEE International Conference on Computer Vision (ICCV)*, 2017, pp. 2980–2988.
- [2] R. K. Mehra, “On the identification of variances and adaptive kalman filtering,” *IEEE Transactions on Automatic Control*, vol. 15, no. 2, pp. 175–184, 1970.